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1 - 38

LIST OF IMPORTANT FORMULAE

(READY RECKNOR)

I. STRAIGHT LINES AND CIRCLES

1. Rectangular Co-ordinates

- (i) The distance between the two points (x_1, y_1) and (x_2, y_2) is

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \text{ or } \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- (ii) The co-ordinates of the point which divides the line joining (x_1, y_1) and (x_2, y_2)

(a) **internally** in the ratio $m_1 : m_2$ are $\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

(b) **externally** in the ratio $m_1 : m_2$ are $\left(\frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2} \right)$.

- (iii) The co-ordinates of the mid-point of the line joining (x_1, y_1) and (x_2, y_2) are

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

2. The Straight line

- (i) General equation of the straight line is $ax + by + c = 0$.

- (ii) (a) Equation of any line parallel to x-axis and at a distance 'k' from it is $y = k$.

(b) Equation of x-axis is $y = 0$.

- (iii) (a) Equation of any line parallel to y-axis and at a distance 'k' from it is $x = k$.

(b) Equation of y-axis is $x = 0$.

- (iv) Equation of a line having slope m and making an intercept 'c' on y-axis is $y = mx + c$.

- (v) Equation of a line whose intercepts on the co-ordinate axes are 'a' and 'b' respectively is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

- (vi) The intercepts of the line $ax + by + c = 0$ are $-\frac{c}{a}, -\frac{c}{b}$.

(vii) Equation of any line passing through the point (x_1, y_1) and having slope 'm' is
 $y - y_1 = m(x - x_1)$.

(viii) Slope of a line joining the two points (x_1, y_1) and (x_2, y_2) is $\frac{y_1 - y_2}{x_1 - x_2}$ or $\frac{y_2 - y_1}{x_2 - x_1}$

i.e., $\frac{\text{difference of ordinates}}{\text{difference of abscissae}}$.

(ix) Slope of a line whose equation is given $= -\frac{\text{coefficient of } x}{\text{coefficient of } y}$

(x) Equation of the line passing through the two points (x_1, y_1) and (x_2, y_2) is $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$.

(xi) The equation of the straight line which is at a distance of p from the origin and $0 \leq \alpha \leq 360^\circ$ is the angle made by the perpendicular with the positive direction of the x-axis measured anti-clockwise is $x \cos \alpha + y \sin \alpha = p$.

(xii) The equation of the line $ax + by + c = 0$ in the Normal form or Perpendicular form is

$$\frac{a}{\sqrt{a^2 + b^2}}x + \frac{b}{\sqrt{a^2 + b^2}}y = \frac{-c}{\sqrt{a^2 + b^2}} \text{ if } c \leq 0; \frac{(-a)}{\sqrt{a^2 + b^2}}x + \frac{(-b)}{\sqrt{a^2 + b^2}}y = \frac{c}{\sqrt{a^2 + b^2}}, \text{ if } c > 0.$$

(xiii) If ' θ ' is an acute angle between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, then

$$\cos \theta = \frac{|a_1a_2 + b_1b_2|}{\sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2}} \text{ and } \tan \theta = \left| \frac{a_1b_2 - a_2b_1}{a_1a_2 + b_1b_2} \right|$$

If θ is an angle between two lines, then $\pi - \theta$ is another angle between the lines.

(xiv) If θ is the acute angle between the lines $y = m_1x + c_1$ and $y = m_2x + c_2$ then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1m_2} \right|.$$

(xiv) Two lines are parallel if $m_1 = m_2$ and perpendicular if $m_1 m_2 = -1$.

(xv) Any line parallel to $ax + by + c = 0$ is $ax + by + k = 0$, where k is a constant.

(xvi) Any line perpendicular to $ax + by + c = 0$ is $bx - ay + k = 0$, where k is a constant.

(xvii) The equation of the line (a) parallel to $ax + by + c = 0$ (b) perpendicular to $ax + by + c = 0$ and passing through (x_1, y_1) is (a) $a(x - x_1) + b(y - y_1) = 0$ (b) $b(x - x_1) - a(y - y_1) = 0$.

(xviii) Length of perpendicular from the point (x_1, y_1) on $ax + by + c = 0$ is $\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$.

(xix) Distance between the parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$.

(xx) The point of intersection of two given lines is obtained by solving them simultaneously.

3. The Circle

- (i) Equation of the **circle** with centre (h, k) and radius ' r ' is $(x - h)^2 + (y - k)^2 = r^2$
- (ii) Equation of the **point circle** with centre (h, k) is $(x - h)^2 + (y - k)^2 = 0$.
- (iii) Equation of the circle with centre origin $(0, 0)$ and radius ' r ' is $x^2 + y^2 = r^2$.
- (iv) The **general equation** of the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$. Its centre $= (-g, -f)$ and radius $= \sqrt{g^2 + f^2 - c}$.
- (v) The centre of the circle $ax^2 + ay^2 + 2gx + 2fy + c = 0$ is $\left(-\frac{g}{a}, -\frac{f}{a}\right)$ and radius is $\frac{1}{a} \cdot \sqrt{g^2 + f^2 - ac}$.
- (vi) Equation of the circle with (x_1, y_1) and (x_2, y_2) as the end points of diameter is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.
- (vii) Let A, B be two points on a circle. Then
- the line $\overline{AB}$ passing through A and B is called a secant line.
 - the line segment $\overline{AB}$ joining A and B is called a chord.
 - AB is called the length of the chord $\overline{AB}$ of the circle.
- (viii) Let P be a point in the circle. Also let Q be any other point in the circle. If the secant line $\overline{PQ}$ approaches the same limiting position as Q moves along the curve and approaches to P from either side, then the limiting position is called a tangent line or tangent to the circle at P. The point P is called point of contact of the tangent to the circle.
- (ix) Radius of the circle is the perpendicular distance from the centre of the circle to the line $ax + by + c = 0$.
- (x) If (x_1, y_1) is one end of a diameter of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, then the other end is $(-2g - x_1, -2f - y_1)$.
- (ix) Equation of tangent at (x_1, y_1) w.r.t the circle
- $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
 - $x^2 + y^2 = a^2$ is $xx_1 + yy_1 = a^2$.
- (x) Equation of the normal at the point (x_1, y_1) w.r.t the circle
- $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(y_1 + f)(x - x_1) - (x_1 + g)(y - y_1) = 0$.
 - $x^2 + y^2 = a^2$ is $y_1x - x_1y = 0$.

4. Conic Sections

- (a) The locus of a point which moves in a plane so that its distance from a fixed point bears a constant ratio to its distance from a fixed straight line is called a conic section or simply conic. The fixed point is called **focus**, the fixed straight line is called **directrix** and the constant ratio 'e' is called **eccentricity** of the conic.
- (b) If $e = 1$, then the conic is called a **parabola**.
 If $e < 1$, then the conic is called an **ellipse**.
 If $e > 1$, then the conic is called a **hyperbola**.
- (c) The equation of a conic is of the form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$.
- (d) A line passing through the focus of a conic is said to be principal axis of the conic if it is perpendicular to the directrix of the conic.
- (e) The points of intersection of a conic and its principal axis are called vertices of the conic.
- (f) A conic has atmost two vertices.
- (g) The midpoint of the line segment joining the vertices of a conic is called centre of the conic.
- (h) A conic is said to be in the standard form if the principal axis of the conic is x-axis and the centre of the conic is the origin.

(i) Parabola

- (a) The standard equation of a parabola is $y^2 = 4ax$.
- (b) For the parabola $y^2 = 4ax$, vertex is $(0, 0)$, focus is $(a, 0)$, equation of the axis is $x = a$ and equation of the directrix is $x = -a$.
- (c) If we rotate the axes 90° in the clockwise direction then the equation $y^2 = 4ax$ of a parabola is transformed to $x^2 = 4ay$.
- (d) For the parabola $x^2 = 4ay$, vertex is $(0, 0)$, focus is $(0, a)$ equation of the latus-rectum is $y = a$ and equation of the directrix is $y = -a$.
- (e) A chord passing through a point P on the parabola and perpendicular to the principal axis of the parabola is called the **double ordinate** of the point P.
- (f) A chord of the parabola passing through the focus is called a **focal chord**.
- (g) A focal chord of a parabola perpendicular to the principal axis of the parabola is called **latus rectum**. If the latus rectum meets the parabola in L and L', then LL' is called length of the latus rectum.
- (h) The length of the latus rectum of the parabola $y^2 = 4ax$ and $x^2 = 4ay$ is $4a$.
- (i) Other forms of the parabola are $y^2 = -4ax$ and $x^2 = -4ay$.

(ii) Ellipse

- (a) The standard form of an equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b$.
- (b) The principal axes of the above ellipse are major axis and minor axis. The equation of the major axis is $y = 0$ and the equation of the minor axis is $x = 0$. The centre of the ellipse is $(0, 0)$.
- (c) The above ellipse has two vertices $A(a, 0)$, $A'(-a, 0)$; two foci $S(ae, 0)$, $S'(-ae, 0)$, two latus rectum $x = \pm ae$ and two directrices $x = \pm \frac{a}{e}$. Its eccentricity is given by $b^2 = a^2(1 - e^2)$.

- (d) The other form of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a < b$.

Its vertices are $A(0, b)$, $A'(0, -b)$; foci are $S(0, be)$, $S'(0, -be)$; two latus-rectum $y = \pm be$ and two directrices $y = \pm \frac{b}{e}$. The equation of major axis is $x = 0$ and minor axis is $y = 0$. Its centre is $(0, 0)$ and eccentricity is $a^2 = b^2(1 - e^2)$.

- (e) The length of the latus rectum of the ellipse

(i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b > 0$) is $\frac{2b^2}{a}$.

(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($0 < a < b$) is $\frac{2a^2}{b}$.

- (f) If P is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci S and S' then $SP + S'P = 2a$.

(iii) Hyperbola

- (a) The equation of a hyperbola in the standard form is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

- (b) It has two vertices $A(a, 0)$, $A'(-a, 0)$; two foci $S(ae, 0)$, $S'(-ae, 0)$, two latus rectum $x = \pm ae$ and two directrices $x = \pm \frac{a}{e}$ and two axes of which one is transverse axis $y = 0$ and the other is conjugate axis $x = 0$.

- (c) Its centre is $(0, 0)$ and eccentricity is given by $b^2 = a^2(e^2 - 1)$ and length of latus rectum is $\frac{2b^2}{a}$.

- (d) The equation of the conjugate hyperbola is $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$. Its centre is (0, 0), vertices are B(0, b), B'(0, -b), foci are S(0, be), S'(0, -be), equations of latus rectum are $y = \pm be$ and directrices are $y = \pm \frac{b}{e}$. The eccentricity is given by $a^2 = b^2(e^2 - 1)$.

Transverse axis is $x = 0$ and conjugate axis is $y = 0$. The length of latus rectum is $\frac{2a^2}{b}$.

- (e) A hyperbola is said to be a rectangular hyperbola if the length of its transverse axis is equal to the length of its conjugate axis. The equation of the rectangular hyperbola is $x^2 - y^2 = a^2$.

- (f) The eccentricity of a rectangular hyperbola is always $\sqrt{2}$.

- (g) If P is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with foci S and S' then $S'P - SP = 2a$.

II. CALCULUS

1. Standard Limits

- (i) $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$, n is any rational number.
- (ii) $\lim_{x \rightarrow a} \frac{x^m - a^m}{x^n - a^n} = \frac{m}{n} a^{m-n}$, m and n are any rational numbers.
- (iii) $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$; $\lim_{\theta \rightarrow 0} \frac{\sin n\theta}{n} = \lim_{\theta \rightarrow 0} \frac{\tan n\theta}{n} = n$
- (iv) $\lim_{\theta \rightarrow 0} \frac{\sin a\theta}{\sin b\theta} = \lim_{\theta \rightarrow 0} \frac{\tan a\theta}{\tan b\theta} = \lim_{\theta \rightarrow 0} \frac{\sin a\theta}{\tan b\theta} = \lim_{\theta \rightarrow 0} \frac{\tan a\theta}{\sin b\theta} = \frac{a}{b}$
- (v) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$
- (vi) $\lim_{n \rightarrow 0} (1 + n)^{1/n} = e$
- (vii) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- (viii) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a$
- (ix) $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$, $\lim_{x \rightarrow 0} \cos\left(\frac{1}{x}\right)$ does not exist.
- (x) $\lim_{x \rightarrow 0} x \cdot \sin\left(\frac{1}{x}\right) = 0$, $\lim_{x \rightarrow 0} x \cdot \cos\left(\frac{1}{x}\right) = 0$.

2. Differentiation - I & II

(i) Let f be a function defined on a neighbourhood of a real number ' a '. Then f is said to be differentiable or derivable at a if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists. The limit is called the derivative or differential coefficient of f at ' a '. It is denoted by $f'(a)$.

(ii) If u, v, w are functions of x , then

$$(a) \quad \frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx} \quad (\text{Product Rule})$$

$$(b) \quad \frac{d}{dx}(Ku) = K \frac{du}{dx} \text{ where } K \text{ is a constant}$$

$$(c) \quad \frac{d}{dx}(c_1 u + c_2 v) = c_1 \frac{du}{dx} + c_2 \frac{dv}{dx} \text{ where } c_1 \text{ and } c_2 \text{ are constants}$$

$$(d) \quad \frac{d}{dx}(uvw) = \frac{du}{dx} vw + u \frac{dv}{dx} w + uv \frac{dw}{dx}$$

$$(e) \quad \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \quad (\text{Quotient Rule})$$

$$(f) \quad \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad (\text{Chain Rule}).$$

$$(iii) \quad \frac{d}{dx}(c) = 0, \text{ where } c \text{ is a constant}$$

$$(iv) \quad \frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(a^x) = a^x \log_e a$$

$$\frac{d}{dx}(\log_e x) = \frac{1}{x}$$

$$(v) \quad \frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx} \{ [f(x)]^n \} = n[f(x)]^{n-1} \cdot f'(x)$$

$$\frac{d}{dx} [e^{f(x)}] = e^{f(x)} \cdot f'(x)$$

$$\frac{d}{dx} [a^{f(x)}] = a^{f(x)} \cdot \log a \cdot f'(x)$$

$$\frac{d}{dx} [\log_e f(x)] = \frac{1}{f(x)} \cdot f'(x)$$

$$\frac{d}{dx} [\sin f(x)] = \cos f(x) \cdot f'(x)$$

$$\frac{d}{dx} [\cos f(x)] = -\sin f(x) \cdot f'(x)$$

$$\frac{d}{dx} [\tan f(x)] = \sec^2 f(x) \cdot f'(x)$$

$$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$$

$$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$$

$$(vi) \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$$

$$(vii) \quad \frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\cosh x) = \sinh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$$

$$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \cdot \tanh x$$

$$\frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \cdot \coth x$$

$$\frac{d}{dx}[\cot f(x)] = -\operatorname{cosec}^2 f(x) \cdot f'(x)$$

$$\frac{d}{dx}[\sec f(x)] = \sec f(x) \cdot \tan f(x) \cdot f'(x)$$

$$\frac{d}{dx}[\operatorname{cosec} f(x)] = -\operatorname{cosec} f(x) \cdot \cot f(x) \cdot f'(x)$$

$$\frac{d}{dx}[\sin^{-1} f(x)] = \frac{1}{\sqrt{1-[f(x)]^2}} \cdot f'(x)$$

$$\frac{d}{dx}[\cos^{-1} f(x)] = \frac{-1}{\sqrt{1-[f(x)]^2}} \cdot f'(x)$$

$$\frac{d}{dx}[\tan^{-1} f(x)] = \frac{1}{1+[f(x)]^2} \cdot f'(x)$$

$$\frac{d}{dx}[\cot^{-1} f(x)] = \frac{-1}{1+[f(x)]^2} \cdot f'(x)$$

$$\frac{d}{dx}[\sec^{-1} f(x)] = \frac{1}{f(x) \cdot \sqrt{[f(x)]^2-1}} \cdot f'(x)$$

$$\frac{d}{dx}[\operatorname{cosec}^{-1} f(x)] = \frac{-1}{f(x) \cdot \sqrt{[f(x)]^2-1}} \cdot f'(x)$$

$$\frac{d}{dx}[\sinh f(x)] = \cosh f(x) \cdot f'(x)$$

$$\frac{d}{dx}[\cosh f(x)] = \sinh f(x) \cdot f'(x)$$

$$\frac{d}{dx}[\tanh f(x)] = \operatorname{sech}^2 f(x) \cdot f'(x)$$

$$\frac{d}{dx}[\coth f(x)] = -\operatorname{cosech}^2 f(x) \cdot f'(x)$$

$$\frac{d}{dx}[\operatorname{sech} f(x)] = -\operatorname{sech} f(x) \cdot \tanh f(x) \cdot f'(x)$$

$$\frac{d}{dx}[\operatorname{cosech} f(x)] = -\operatorname{cosech} f(x) \cdot \coth f(x) \cdot f'(x)$$

(viii) If x and y are functions of t then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

(ix) Derivative of $f(x)$ with respect to $g(x)$ is $\frac{f'(x)}{g'(x)}$.

(x) If $y = \sqrt{f(x)} + \sqrt{f(x)} + \sqrt{f(x)} + \dots \infty$ (or) if $f(x) = \sqrt{f(x) + y}$ then $\frac{dy}{dx} = \frac{f'(x)}{2y-1}$.

(xi) A function $f(x, y)$ is said to be homogeneous function of degree 'n' if $f(\lambda x, \lambda y) = \lambda^n f(x, y)$ for all values of λ and n is a constant.

(xii) **Euler's Theorem** : If $f(x, y)$ is a homogeneous function of degree 'n' then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf$.

3. Applications of Derivatives

(i) (a) Equation of tangent to the curve $y = f(x)$ at (x_1, y_1) is

$$y - y_1 = m(x - x_1) \text{ where } m = \left(\frac{dy}{dx} \right)_{(x_1, y_1)}$$

(b) Equation of normal to the curve $y = f(x)$ at (x_1, y_1) is

$$y - y_1 = -\frac{1}{m}(x - x_1) \text{ where } m = \left(\frac{dy}{dx} \right)_{(x_1, y_1)}$$

(c) Lengths of the tangent, normal, subtangent and subnormal to the curve $y = f(x)$ at (x_1, y_1) respectively are

$$\left| \frac{y_1}{m} \cdot \sqrt{1+m^2} \right|, \left| y_1 \cdot \sqrt{1+m^2} \right|, \left| \frac{y_1}{m} \right| \text{ and } |y_1 \cdot m| \text{ where } m = \left(\frac{dy}{dx} \right)_{(x_1, y_1)}.$$

(d) If θ is the acute angle between two curves intersecting at (x_1, y_1) then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \text{ where } m_1 = \left(\frac{dy}{dx} \right)_{(x_1, y_1)} \text{ of the first curve and } m_2 = \left(\frac{dy}{dx} \right)_{(x_1, y_1)}$$

of the second curve.

(e) Two curves are said to be orthogonal if $m_1 m_2 = -1$ and the two curves touch each other if $m_1 = m_2$.

(ii) Maxima and Minima :

(a) Let $f : A \rightarrow R$ be a function. Then

- (i) f is said to be monotonically increasing on A if $x_1, x_2 \in A, x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$
- (ii) f is said to be monotonically decreasing on A if $x_1, x_2 \in A, x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$
- (iii) f is said to be monotonic on A if f is either monotonically increasing or monotonically decreasing on A .

(b) The function $y = f(x)$ has a maximum at $x = a$ if $f'(a) = 0$ and $f''(a) < 0$ i.e., $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$ at $x = a$. The maximum value of $f(x)$ is $f(a)$.

(c) The function $y = f(x)$ has a minimum at $x = a$ if $f'(a) = 0$ and $f''(a) > 0$ i.e., $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$ at $x = a$. The minimum value of $f(x)$ is $f(a)$.

STRAIGHT LINES AND CIRCLES

UNIT I

Chapter-1

Straight Lines

1.1 - 1.31

Chapter-2

Circle

2.1 - 2.31

CHAPTER

1

STRAIGHT LINES

"Mathematics in India invariably makes one to think of one extraordinary figure- This was Srinivasa Ramanujan."

– Jawahar Lal Nehru

1.1 INTRODUCTION

In earlier classes, we have learnt the coordinates of a point in a plane with reference to two mutually perpendicular lines in the plane. We have also learnt about finding distance between two points, slope of a line and various forms of a straight line, distance between two parallel lines and angle between two intersecting lines.

1.1.1 Two Dimensional Coordinate System

The study of Geometry by means of Algebra was first introduced by Rene Descartes (1596-1650), a French Mathematician. This study is known as Analytical Geometry (also called Co-ordinate Geometry or Algebraic Geometry).

The basic concept in co-ordinate geometry lies in the fact that a point in a plane can be fixed if two properties of the point are known. These are called the co-ordinates of the point.

1.1.2 Rectangular Cartesian Co-ordinate System

Let $X'OX$ and $Y'OY$ be two straight lines at right angles to each other in the plane of the paper. These lines $X'OX$ and $Y'OY$ are called the x -axis and y -axis respectively and they are together called the axes of coordinates or briefly the axes. The point of intersection of the two axes is called the origin and it is denoted by the letter O .

Let P be any point in the plane. Draw PL perpendicular to the x -axis. The lengths OL and LP are called the coordinates of the point P . OL is called the x -coordinate or *abscissa* and LP is called y -coordinate or *ordinate* of the point P . The *abscissa* and the *ordinate* together are called the **co-ordinates** of the point. If $OL = x$ and $LP = y$ then the coordinates of P are denoted by the symbol (x, y) .

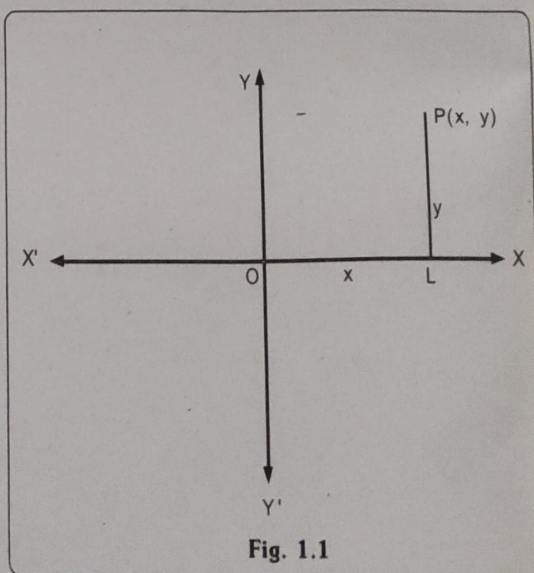


Fig. 1.1

1.1.3 Signs of Coordinates

The two lines $X'OX$ and $Y'OY$ divide the plane into four parts called quadrants. XOY , YOX' , $X'OY'$ and $Y'OX$ are respectively the first, second, third and fourth quadrants. It is a convention to take the lengths measured in the direction of OX and OY as positive and those measured in opposite directions, OX' and OY' as negative. The signs of the coordinates are marked in the Fig.1.2.

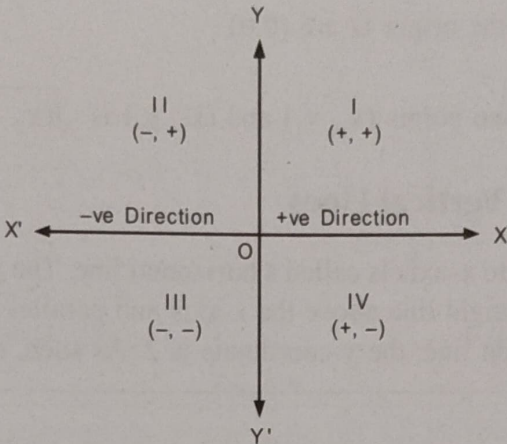


Fig. 1.2

When the axes are taken at right angles as described above, it is called a system of Rectangular cartesian coordinates. The following table gives the sign convention of x and y , the coordinates of a point in several quadrants.

	x	y
I Quadrant (XOY)	$+$	$+$
II Quadrant ($X'OY$)	$-$	$+$
III Quadrant ($X'OY'$)	$-$	$-$
IV Quadrant ($Y'OX$)	$+$	$-$

Thus we can determine the position of a point, if we know the coordinates with their proper signs. For example, to plot the point $(-3,4)$ we measure a length OM' 3 units along OX' and MQ' 4 units along a line perpendicular to OX' in the direction of OY .

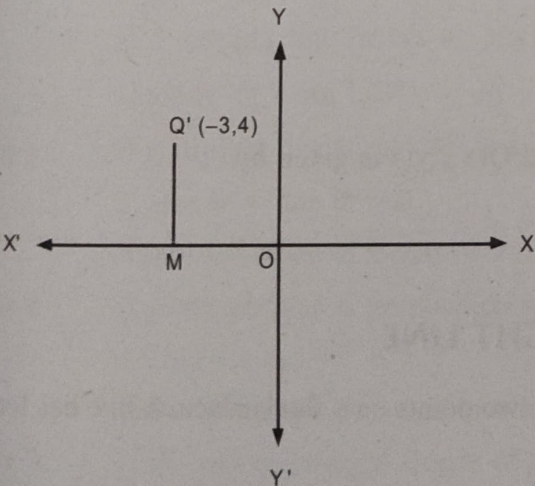


Fig. 1.3

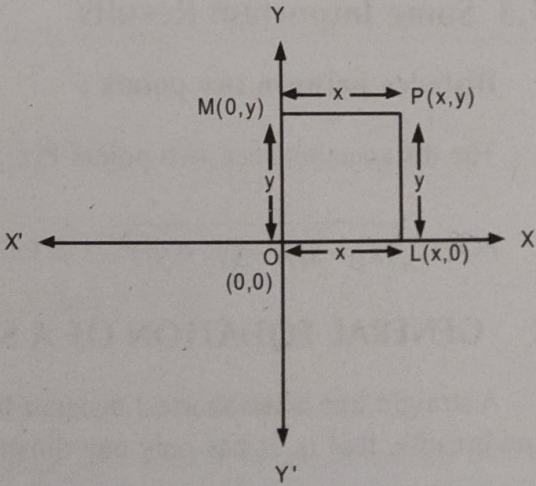


Fig. 1.4

Note 1 : The x-coordinate of P is the distance of P from the y-axis and y-coordinate is its distance from the x-axis.

Note 2 : If a point lies on the x-axis, its y-coordinate is zero and if the point lies on y-axis, then its x-coordinate is zero. Thus any point on the x-axis is of the form $(x, 0)$ and any point on the y-axis is of the form $(0, y)$.

Note 3 : The coordinates of the origin O are $(0, 0)$.

Note 4 : Distance between two points (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

1.1.4 Horizontal and Vertical Lines

A straight line parallel to x-axis is called a horizontal line. The y-coordinate for any point on x-axis is zero. Consider a straight line above the x-axis and parallel to it at a distance of 2 units. For every point on this straight line, the y-coordinate is 2. As such, equation of the line is $y = 2$.

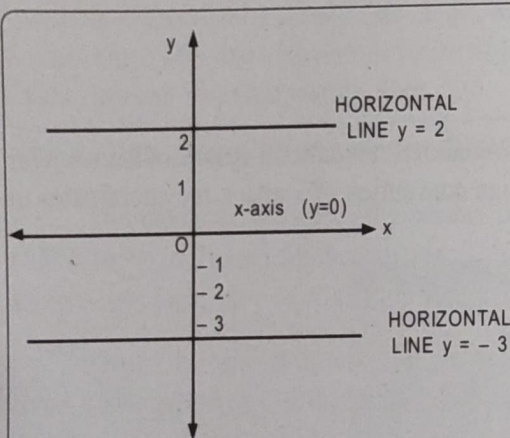


Fig. 1.4

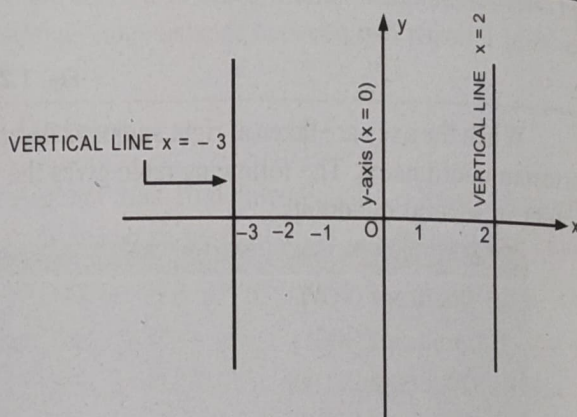


Fig. 1.5

Any line parallel to y-axis or perpendicular to x-axis is called a vertical line. The equation of any straight line parallel to y-axis is of the form $x = k$, where k is a real number.

1.1.5 Some Important Results

Distance between two points :

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

1.2 GENERAL EQUATION OF A STRAIGHT LINE

A straight line is the shortest distance between two points on a flat surface. A line has length but no breadth, that is, it has only one dimension.

An equation of the form $ax + by + c = 0$, where a, b, c are real numbers, not all zero, always represents a straight line.

1.3 SLOPE OF A STRAIGHT LINE

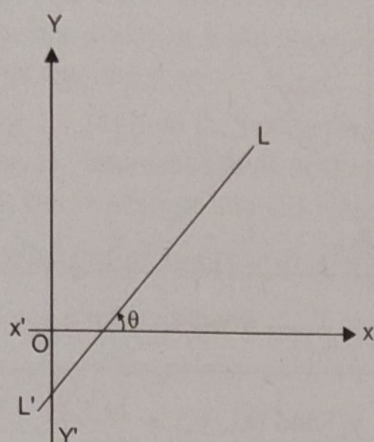
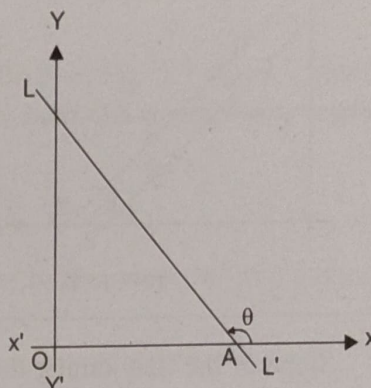

 $\theta \text{ acute} \Rightarrow \tan \theta > 0$

 $\theta \text{ obtuse} \Rightarrow \tan \theta < 0$

Fig. 1.6

Let a straight line $L'L$ intersect the x -axis at A . Let AX be revolved about the point A in the anti-clock wise (+ve) direction, until it coincides with $L'L$. The angle through which AX has been turned in doing so is known as the inclination of the straight line $L'L$ and is generally denoted by ' θ '. This angle can have any value from 0° to 180° .

The tangent of the angle of inclination of the straight line is called **slope** or **gradient** of the straight line and is usually denoted by ' m ', so that $m = \tan \theta$.

' m ' is positive or negative according as θ is acute ($0^\circ \leq \theta < 90^\circ$) or obtuse ($90^\circ < \theta \leq 180^\circ$).

Definition :

If a non-vertical straight line makes an angle θ with the x -axis measured anticlock wise from the positive direction of the x -axis, then $\tan \theta$ is called the **slope** or **gradient** of the straight line.

Alternatively, if θ is the inclination of a line then $\tan \theta$ is called the slope or gradient of the line l .

For example :

- (1) If a straight line makes an angle 45° with the positive direction of the x -axis, then its slope is given by $m = \tan \theta = \tan 45^\circ = 1$
- (2) If a straight line makes an angle 120° with the positive direction of the x -axis, its slope is $m = \tan 120^\circ = -\sqrt{3}$

Note 1 : If a straight line is parallel to x -axis, then $\theta = 0^\circ$

$$\therefore \tan \theta = \tan 0^\circ = 0$$

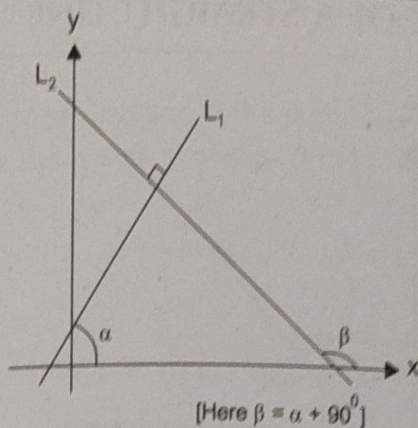
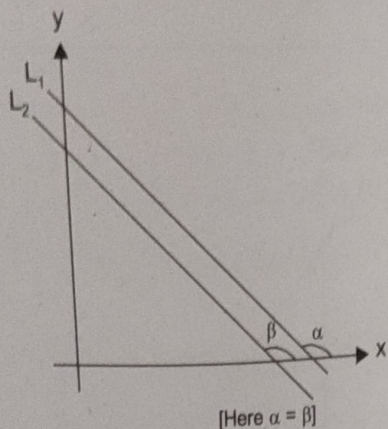
Hence the slope of x -axis or a line parallel to x -axis is zero.

Note 2 : If a straight line is perpendicular to x -axis then $\theta = 90^\circ$

$$\therefore \tan \theta = \tan 90^\circ = \infty$$

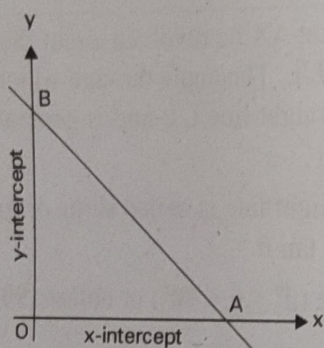
Hence the slope of y -axis or a line parallel to y -axis is **infinite**.

Note 3 : If m_1 and m_2 are the slopes of two parallel lines then $m_1 = m_2$ (since the two lines inclined at the same angle θ to the x -axis) and $m_1 \cdot m_2 = -1$ if the two lines are perpendicular.

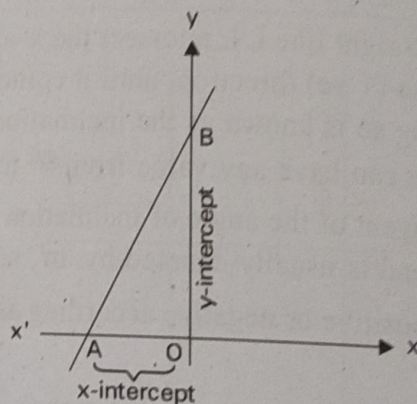


Note 4 : Slope of the line joining the two points (x_1, y_1) and (x_2, y_2) is $\frac{y_2 - y_1}{x_2 - x_1}$.

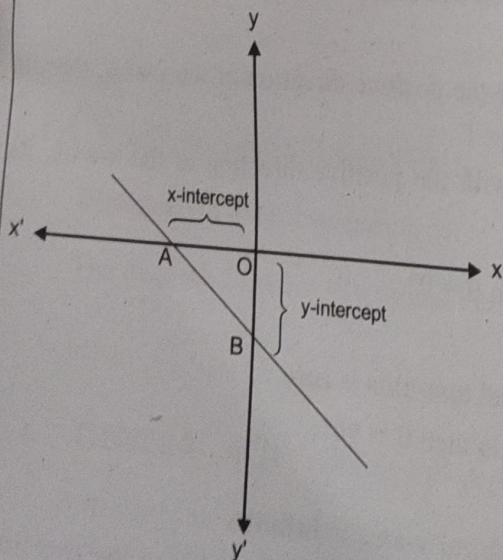
1.4 INTERCEPTS ON CO-ORDINATE AXES



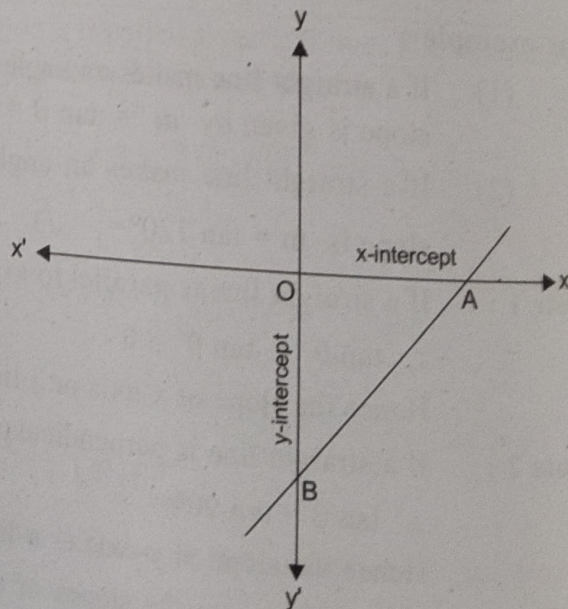
(a)



(b)



(c)



(d)

Fig. 1.7

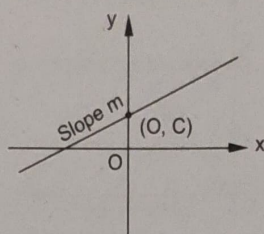
Definition :

A part of a line or plane cut off by another line or plane is called Intercept. If a straight line AB intersects the x-axis in A and y-axis in B, then OA is called the x-intercept and OB is called the y-intercept of the line.

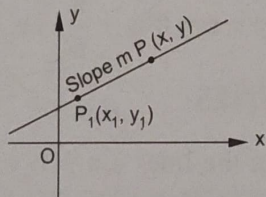
In Fig. 1.7 (a) both the intercepts OA, OB are positive. In Fig. 1.7 (b), x - intercept OA is negative and y - intercept OB is positive. In Fig. 1.7 (c) both OA and OB are negative and in Fig. 1.7 (d), OA is positive and OB is negative.

1.5 VARIOUS FORMS OF A STRAIGHT LINE

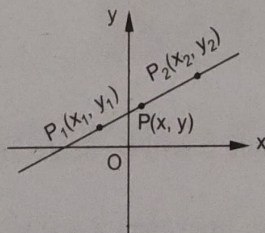
- (1) **Slope-Intercept Form :** Equation of the line having slope 'm' and cutting an intercept 'c' from the y-axis is $y = mx + c$.



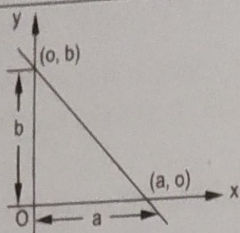
- (2) **Point-Slope Form :** Equation of the line passing through the point (x_1, y_1) and having slope 'm' is $y - y_1 = m(x - x_1)$.



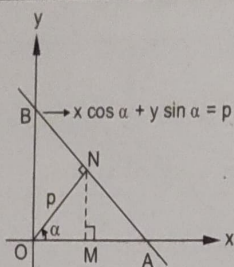
- (3) **Two-Point Form :** Equation of the line passing through the points (x_1, y_1) and (x_2, y_2) is $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$.



- (4) **Intercept Form :** Equation of a line which makes intercepts 'a' with x-axis and 'b' with y-axis is $\frac{x}{a} + \frac{y}{b} = 1$.



- (5) **Normal Form or Perpendicular Form :** The equation of the straight line, whose distance from the origin is 'p', ($p > 0$) and α ($0 \leq \alpha < 360^\circ$) is the angle which normal makes with the positive direction of the x-axis measured anti-clock-wise is $x \cos \alpha + y \sin \alpha = p$. This is known as normal form.



- (6) **General Equation of a Line :** An equation of first degree in x and y represents a straight line. So an equation of the form $ax + by + c = 0$ where a, b, c are real numbers will always represents a straight line.

Note : Slope of the line $ax + by + c = 0$ is $-\frac{a}{b}$ i.e., $-\frac{\text{coefficient of } x}{\text{coefficient of } y}$.

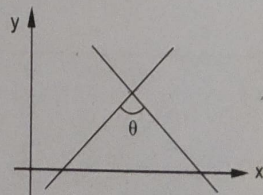
1.6 ANGLE BETWEEN TWO LINES

- (i) If ' θ ' is an acute angle between the lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, then

$$\cos \theta = \frac{|a_1a_2 + b_1b_2|}{\sqrt{a_1^2 + b_1^2} \times \sqrt{a_2^2 + b_2^2}} \quad \text{and} \quad \tan \theta = \frac{|a_1b_2 - a_2b_1|}{|a_1a_2 + b_1b_2|}$$

- (ii) If m_1 and m_2 are the slopes of two lines and if θ is the angle between the lines, then

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$$



Note 1 : If $\frac{m_1 - m_2}{1 + m_1 m_2}$ is positive, the angle θ will be acute and if negative, θ will be obtuse.

Note 2 : If θ is an angle between the two lines then $\pi - \theta$ is another angle between the lines.

Cor. 1 : Condition for Parallelism. Two straight lines will be parallel, if their slopes are equal i.e., if $m_1 = m_2$.

Cor. 2 : Condition for Perpendicularity. Two straight lines will be perpendicular to each other, if the product of the slopes of the two lines is -1 i.e., if $m_1 \cdot m_2 = -1$.

1.7 POINT OF INTERSECTION OF TWO LINES

Theorem :

The point of intersection of two non-parallel lines $a_1x + b_1y + c_1 = 0$ and

$$a_2x + b_2y + c_2 = 0 \text{ is } \left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right), \text{ provided } a_1b_2 \neq a_2b_1$$

Proof :

Let (α, β) be the point of intersection of the lines

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0.$$

Then the point (α, β) lies on both the lines.

$$\therefore a_1\alpha + b_1\beta + c_1 = 0$$

$$a_2\alpha + b_2\beta + c_2 = 0$$

Solving by cross-multiplication,

$$\frac{\alpha}{b_1c_2 - b_2c_1} = \frac{\beta}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\Rightarrow \alpha = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \beta = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

$\therefore$ The point of intersection is

$$\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right) [\because a_1b_2 - a_2b_1 \neq 0 \text{ as the lines are not parallel}]$$

Note : The point of intersection of two lines is obtained by solving them simultaneously.

1.8 EQUATION TO A LINE PARALLEL AND PERPENDICULAR TO $AX + BY + C = 0$

(i) Equation of any line **parallel to** $ax + by + c = 0$ is of the form $ax + by + k = 0$, where k is a constant.

Thus equations of two parallel straight lines differ only in constant terms.

(ii) Equation of any line **perpendicular to** $ax + by + c = 0$ is of the form $bx - ay + k = 0$, where k is a constant.

Note : 1. Equation of a line parallel to $ax + by + c = 0$ and passing through the point (x_1, y_1) is $a(x - x_1) + b(y - y_1) = 0$.

2. Equation of a line perpendicular to $ax + by + c = 0$ and passing through the point (x_1, y_1) is $b(x - x_1) - a(y - y_1) = 0$.

1.9 DISTANCE OF A POINT FROM A LINE

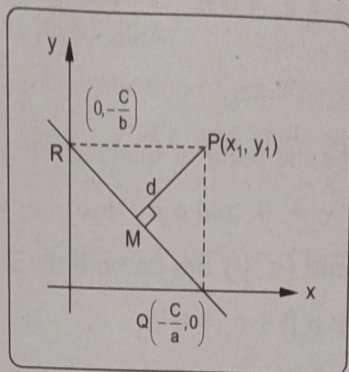
The distance of a point from a line is the length of the perpendicular drawn from the point to the line.

Thus, the perpendicular distance of a line $ax + by + c = 0$ from a point (x_1, y_1) is given by

$$\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

[Hint : area of $\Delta PQR = \frac{1}{2} PM \cdot QR$

$$\Rightarrow PM = \frac{2(\text{area of } \Delta PQR)}{QR}]$$

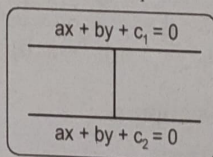


1.10 DISTANCE BETWEEN TWO PARALLEL LINES

Distance between two lines is equal to the length of the perpendicular from a point on line (1) to line (2).

The distance between two parallel straight lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is

$$\frac{|c_2 - c_1|}{\sqrt{a^2 + b^2}}$$



Note : While finding the distance between the parallel lines, the coefficients of x and y in the two equations must be made equal.

Solved Examples

Example 1:

Find the slope of the line

- passing through the points $(3, -2)$ and $(3, 4)$
- passing through the points $(3, -2)$ and $(-1, 4)$
- making inclination of 60° with the positive direction of x -axis.

Solution :

- (i) We are given $(x_1, y_1) = (3, -2)$ and $(x_2, y_2) = (3, 4)$.

$\therefore$ The slope of the line through $(3, -2)$ and $(3, 4)$ is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - (-2)}{3 - 3} = \frac{6}{0} = \infty \text{ (infinite), which is not defined.}$$

(ii) Here $(x_1, y_1) = (3, -2)$ and $(x_2, y_2) = (-1, 4)$

$$\therefore \text{Slope, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - (-2)}{-1 - 3} = \frac{6}{-4} = -\frac{3}{2}.$$

(iii) Given inclination of the line, $\theta = 60^\circ$

$$\therefore \text{Slope of the line} = \tan \theta = \tan 60^\circ = \sqrt{3}.$$

Example 2 :

Find the equation of the straight line passing through the point

(i) $(1, 2)$ and having slope 5 (ii) $(1, 1)$ with slope -1

(iii) $(-2, 5)$ and having slope $-\frac{3}{4}$.

[Jan. 2023 (MID-I)]

Solution :

(i) Given $(x_1, y_1) = (1, 2)$

and slope, $m = 5$

$\therefore$ The equation of the line is

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - 2 = 5(x - 1) \Rightarrow y - 2 = 5x - 5$$

$$\Rightarrow 5x - y - 3 = 0$$

(ii) Here $(x_1, y_1) = (1, 1)$ and slope, $m = -1$

The equation of the required line is $y - 1 = -1(x - 1) \Rightarrow y - 1 = -x + 1 \Rightarrow x + y - 2 = 0$

(iii) Equation of the required line is $y - y_1 = m(x - x_1)$

$$\text{i.e., } y - 5 = -\frac{3}{4}(x + 2)$$

$$\text{i.e., } 4(y - 5) = -3(x + 2)$$

$$\text{i.e., } 4y - 20 = -3x - 6$$

$$\text{or } 3x + 4y - 14 = 0$$

Example 3 :

Find the slope of a line passing through the points $A(3, -7)$ and $B(2, -5)$ and also find equation.

Solution :

We are given

$$A(3, -7) = A(x_1, y_1) \Rightarrow x_1 = 3, y_1 = -7$$

(ii) Here $(x_1, y_1) = (3, -2)$ and $(x_2, y_2) = (-1, 4)$

$$\therefore \text{Slope, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - (-2)}{-1 - 3} = \frac{6}{-4} = -\frac{3}{2}.$$

(iii) Given inclination of the line, $\theta = 60^\circ$

$$\therefore \text{Slope of the line} = \tan \theta = \tan 60^\circ = \sqrt{3}.$$

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[Jan. 2023 (MID-I)]

Solution :

(i) Given $(x_1, y_1) = (1, 2)$

and slope, $m = 5$

$\therefore$ The equation of the line is

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - 2 = 5(x - 1) \Rightarrow y - 2 = 5x - 5$$

$$\Rightarrow 5x - y - 3 = 0$$

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The equation of the required line is $y - 1 = -1(x - 1) \Rightarrow y - 1 = -x + 1 \Rightarrow x + y - 2 = 0$.

(iii) Equation of the required line is $y - y_1 = m(x - x_1)$

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$$\text{or } 3x + 4y - 14 = 0$$

Example 3 :

Find the slope of a line passing through the points $A(3, -7)$ and $B(2, -5)$ and also find its equation.

Solution :

We are given

$$A(3, -7) = A(x_1, y_1) \Rightarrow x_1 = 3, y_1 = -7$$

$$\text{and } B(2, -5) = B(x_2, y_2) \Rightarrow x_2 = 2, y_2 = -5$$

$$\text{Now slope of the line AB, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-5 - (-7)}{2 - 3} = \frac{-5 + 7}{-1} = \frac{2}{-1} = -2$$

We can find equation of the required line passing through either the point A or the point B.

$\therefore$ Equation of the line AB passing through the point A(3, -7) is given by

$$y - y_1 = m(x - x_1) \Rightarrow y - (-7) = (-2)(x - 3)$$

$$\Rightarrow y + 7 = -2x + 6 \Rightarrow 2x + y + 1 = 0$$

Observation : Equation of the line AB passing through the point B(2, -5) is also given by

$$y - (-5) = (-2)(x - 2) \Rightarrow y + 5 = -2x + 4 \Rightarrow 2x + y + 1 = 0.$$

Example 4:

Find the equation of the line passing through the point (i) (3, -4) and having inclination 60° (ii) (1, 3) and having inclination 30° . [Jan. 2023 (MID-I)]

Solution :

(i) We are given $(x_1, y_1) = (3, -4)$ and inclination, $\theta = 60^\circ$.

$$\therefore \text{Slope of the line, } m = \tan \theta = \tan 60^\circ = \sqrt{3}$$

$$\text{Hence equation of the line is } y - y_1 = m(x - x_1)$$

$$\text{i.e., } y - (-4) = \sqrt{3}(x - 3) \text{ (or) } y + 4 = \sqrt{3}x - 3\sqrt{3} \text{ (or) } \sqrt{3}x - y = 4 + 3\sqrt{3}.$$

(ii) Given $(x_1, y_1) = (1, 3)$, and $\theta = 30^\circ$. So $m = \tan 30^\circ = \frac{1}{\sqrt{3}}$

$$\therefore \text{Equation of the required line is } y - 3 = \frac{1}{\sqrt{3}}(x - 1) \Rightarrow \sqrt{3}(y - 3) = x - 1 \Rightarrow x - \sqrt{3}y + 3\sqrt{3} - 1 = 0$$

Example 5:

Find the equation of the straight line passing through the points

(i) (0, 1) and (3, -4). [April 2018 (D.E.C.E)]

(ii) (-5, 2) and (3, -2) [Oct. 2015 (D.E.C.E), April 2018 (D.C.M.E)]

(iii) (1, -2) and (-2, 1). [April 2018 (D.C.E)]

(iv) (1, 1) and (0, 2).

Solution :

(i) We are given $(x_1, y_1) = (0, 1)$ and $(x_2, y_2) = (3, -4)$

$\therefore$ The required equation of the straight line is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\text{i.e., } \frac{y-1}{-4-1} = \frac{x-0}{3-0} \Rightarrow \frac{y-1}{-5} = \frac{x}{3}$$

$$\Rightarrow 3(y-1) = -5x \Rightarrow 3y-3 = -5x$$

$$\Rightarrow 5x + 3y - 3 = 0$$

(ii) This is left as an exercise to the reader. Proceeding as in (i), we get the equation of the required line as $x + 2y + 1 = 0$.

(iii), (iv) This is left as an exercise to the reader.

Example 6:

(i) Equation of a line is $3x - 4y + 10 = 0$. Find its slope and y-intercept.

(ii) Find the slope of the line $3x - 2y + 1 = 0$.

(iii) If the slope of the straight line $kx + 2y + 3 = 0$ is 2, then find k .

[Jan. 2023 (MID-I)]

Solution :

(i) Given line can be written as

$$\Rightarrow 4y = 3x + 10$$

$$\Rightarrow y = \frac{3}{4}x + \frac{10}{4} = \frac{3}{4}x + \frac{5}{2} \quad \dots (1)$$

Comparing (1) with $y = mx + c$, we have slope of the given line, $m = \frac{3}{4}$ and y-intercept = $\frac{5}{2}$.

(ii) We know that the slope of the line $ax + by + c = 0$ is $\frac{-a}{b}$ i.e., $\frac{-\text{coefficient of } x}{\text{coefficient of } y}$

$\therefore$ Slope of the line $3x - 4y + 10 = 0$ is $\frac{-3}{-4}$ i.e., $\frac{3}{4}$.

(iii) Slope of the given line = $\frac{-a}{b} = \frac{-\text{coefficient of } x}{\text{coefficient of } y} = \frac{-k}{2} \quad \dots (1)$

But Given slope of the given line = 2

From (1) and (2), we get $\dots (2)$

$$\frac{-k}{2} = 2 \quad \therefore k = -4.$$

Example 7:

Find the equation of the straight line making intercepts

(i) $\frac{1}{3}$ and $\frac{4}{5}$

(ii) $\frac{15}{3}$ and $\frac{7}{5}$ [April 2013]

(iii) -4 and 1

with the x -axis and y -axis respectively.

Solution :

(i) We are given

$$x\text{-intercept} = a = \frac{1}{3} \text{ and } y\text{-intercept} = b = \frac{4}{5}$$

$$\therefore \text{The equation of the required straight line is } \frac{x}{a} + \frac{y}{b} = 1$$

$$\text{i.e., } \frac{x}{1/3} + \frac{y}{4/5} = 1 \text{ or } 3x + \frac{5y}{4} = 1 \text{ or } \frac{12x + 5y}{4} = 1$$

$$\text{or } 12x + 5y - 4 = 0$$

(ii) This is left as an exercise to the student. Proceeding as in (i), we get the equation of the required line as $21x + 75y - 105 = 0$. (ii)

(iii) This is left as an exercise to the reader.

Example 8:

Write the intercept form of straight line $3x - 2y - 2 = 0$.

[April 2017 (D.M.E)]

Solution :

The given line can be rewritten as $3x - 2y = 2$. Divide by 2. We get

$$\frac{3x}{2} - \frac{2y}{2} = 1$$

$$\text{i.e., } \frac{x}{2/3} - \frac{y}{1} = 1$$

$$\text{or } \frac{x}{2/3} + \frac{y}{(-1)} = 1$$

which is the intercept form of given straight line.

Example 9:

Find the intercepts made by the straight line

(i) $13x + 7y + 11 = 0$

(ii) $2x + 3y - 6 = 0$

(iii) $3x + 2y - 2 = 0$

on the coordinate axes.

Solution :

- (i) We know that the intercept form of a straight line $ax + by + c = 0$ is $\frac{x}{a} + \frac{y}{b} = 1$.

So we rewrite the given straight line as

$$13x + 7y = -11.$$

Now divide on both sides by -11 . We get

$$\frac{13x}{-11} + \frac{7y}{-11} = 1.$$

This can be written as $\frac{x}{-11/3} + \frac{y}{-11/7} = 1$ which is of the form $\frac{x}{a} + \frac{y}{b} = 1$.

Hence the x-intercept is $\frac{-11}{3}$ and y-intercept is $\frac{-11}{7}$.

- (ii) The given straight line can be written as

$$2x + 3y = 6$$

Now divide on both sides by 6. Then we get $\frac{2x}{6} + \frac{3y}{6} = 1$

i.e., $\frac{x}{3} + \frac{y}{2} = 1$ which is of the form $\frac{x}{a} + \frac{y}{b} = 1$.

$\therefore$ x - intercept = 3 and y - intercept = 2.

- (iii) This is left as an exercise to the reader.

Example 10:

Find the angle between the two straight lines $2x + y + 3 = 0$ and $x + y - 2 = 0$. [May 2022]

Solution :

Method 1 : We have

$$a_1 = 2, b_1 = 1, c_1 = 3 \text{ and } a_2 = 1, b_2 = 1, c_2 = -2$$

We know that the acute angle 'θ' between the straight lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is given by

$$\cos \theta = \frac{|a_1 a_2 + b_1 b_2|}{\sqrt{a_1^2 + b_1^2} \times \sqrt{a_2^2 + b_2^2}} \quad (\text{or}) \quad \tan \theta = \left| \frac{a_1 b_2 - a_2 b_1}{a_1 a_2 + b_1 b_2} \right|$$

$\therefore$ The angle θ between the given lines is given by

$$\cos \theta = \frac{(2)(1) + (1)(1)}{\sqrt{(2)^2 + (1)^2} \cdot \sqrt{(1)^2 + (1)^2}} = \frac{3}{\sqrt{5} \cdot \sqrt{2}} = \frac{3}{\sqrt{10}} \quad \text{So } \theta = \cos^{-1} \left(\frac{3}{\sqrt{10}} \right)$$

$$(\text{or}) \quad \tan \theta = \left| \frac{a_1 b_2 - a_2 b_1}{a_1 a_2 + b_1 b_2} \right| = \frac{(2)(1) - (1)(1)}{(2)(1) + (1)(1)} = \frac{2-1}{2+1} = \frac{1}{3}$$

$$\therefore \theta = \tan^{-1} \left(\frac{1}{3} \right)$$

Method 2 : Using Slopes of the lines

We know that the slope of the straight line $ax + by + c$ is $-\frac{a}{b}$ (i.e., $-\frac{\text{coeff. of } x}{\text{coeff. of } y}$)

$$\therefore \text{ Slope of the line } 2x + y + 3 = 0 \text{ is } m_1 = \frac{-2}{1} = -2$$

$$\text{and slope of the line } x + y - 2 = 0 \text{ is } m_2 = \frac{-1}{1} = -1$$

Hence the acute angle θ between the given lines is given by

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{-2 - (-1)}{1 + (-2)(-1)} \right| = \left| \frac{-2+1}{1+2} \right| = \left| \frac{-1}{3} \right| = \frac{1}{3}$$

$$\therefore \theta = \tan^{-1} \left(\frac{1}{3} \right) = \left| \frac{-2+1}{1+2} \right| = \frac{1}{3}$$

Example 11:

Find the angle between the lines $y - \sqrt{3}x - 5 = 0$ and $\sqrt{3}y - x + 6 = 0$.

[Sept. 2021]

Solution :

Given lines can be written as

$$y = \sqrt{3}x + 5 \quad \dots (1)$$

$$\sqrt{3}y = x - 6 \quad \text{or} \quad y = \frac{1}{\sqrt{3}}x - 2\sqrt{3} \quad \dots (2)$$

The lines (1) and (2) are in the form $y = mx + c$

Thus slope of line (1) is $m_1 = \sqrt{3}$ and slope of line (2) is $m_2 = \frac{1}{\sqrt{3}}$

Let θ be the acute angle between the given lines. Then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{1 + \sqrt{3} \cdot \frac{1}{\sqrt{3}}} \right| = \frac{2/\sqrt{3}}{2} = \frac{1}{\sqrt{3}}$$

$$\therefore \theta = 30^\circ$$

Hence, angle between two lines is either 30° or $180^\circ - \theta$ i.e., $180^\circ - 30^\circ = 150^\circ$.

Alternate Method : Refer Solved Example 10.

Example 12:

Find the point of intersection of the lines $3x + 4y + 6 = 0$ and $6x + 5y - 9 = 0$.

Solution :

The point of intersection of the lines

$$3x + 4y + 6 = 0 \quad \dots (1)$$

$$6x + 5y - 9 = 0 \quad \dots (2)$$

is obtained by solving their equations.

Solving (1) and (2) by cross - multiplication, we get

$$\frac{x}{-36 - 30} = \frac{-y}{-27 - 36} = \frac{1}{15 - 24} \Rightarrow \frac{x}{-66} = \frac{y}{63} = \frac{-1}{9}$$

$$\therefore x = \frac{66}{9} = \frac{22}{3}, y = \frac{-63}{9} = -7$$

Hence the point of intersection is $\left(\frac{22}{3}, -7\right)$.

Another Method : Using Ordinary Elimination Process.

The given lines can be written as

$$3x + 4y = -6 \quad \dots (1)$$

$$6x + 5y = 9 \quad \dots (2)$$

To find one unknown (either x or y), we have to eliminate the other one. Suppose we want to eliminate x . For this, we have to make the coefficients of x must be equal.

$$(1) \times 2 - (2) \text{ gives}$$

$$3y = -21 \Rightarrow y = -7$$

Now we can find the other unknown x by substituting the value of y in (1) or (2).

Substituting the value of y in (1), we get

$$3x + 4(-7) = -6 \Rightarrow 3x - 28 = -6 \Rightarrow 3x = -6 + 28 = 22 \Rightarrow x = \frac{22}{3}$$

$\therefore$ The point of intersection is $\left(\frac{22}{3}, -7\right)$.

Example 13:

Find the perpendicular distance of the point

(i) $(-2, 3)$ from the line $2x - y - 3 = 0$.

[Oct. 2015 (D.C.E)]

(ii) $(1, 2)$ from the line $3x - 4y + 5 = 0$.

[April 2015 (D.E.E.E, D.C.E)]

(iii) $(3, 2)$ from the line $4x + 5y + 6 = 0$.

(iv) $(2, 3)$ from the line $2x - 3y - 5 = 0$.

Solution :

(i) Here $(x_1, y_1) = (-2, 3)$ and $a = 2, b = -1, c = -3$

The perpendicular distance of the given point from given line is

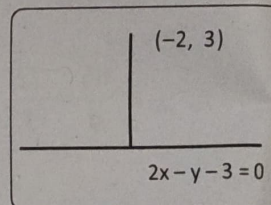
$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|2(-2) - 1(3) + (-3)|}{\sqrt{2^2 + (-1)^2}} = \frac{|-4 - 3 - 3|}{\sqrt{4 + 1}} = \frac{10}{\sqrt{5}} = 2\sqrt{5}.$$

(ii) This is left as an exercise to the reader.

(iii) Proceeding as above, we get the required distance $= d = \frac{28}{\sqrt{41}}$

(iv) Here $(x_1, y_1) = (2, 3)$ and $a = 2, b = -3, c = -5$

$$\begin{aligned} \therefore \text{Required distance} &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|2(2) - 3(3) - 5|}{\sqrt{2^2 + 3^2}} = \frac{|4 - 9 - 5|}{\sqrt{4 + 9}} \\ &= \frac{|-10|}{\sqrt{13}} = \frac{10}{\sqrt{13}} \end{aligned}$$



Example 14:

Find the distance from the origin to the line $3x + 4y + 5 = 0$.

[April 2016 (A/AA/AEI...)]

Solution :

Here $(x_1, y_1) = (0, 0)$ and $a = 3, b = 4, c = 5$

$$\therefore \text{Required distance} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3(0) + 4(0) + 5|}{\sqrt{3^2 + 4^2}} = \frac{5}{\sqrt{25}} = \frac{5}{5} = 1.$$

Example 15:

Find the distance between parallel lines

(i) $5x + 12y - 7 = 0$ and $5x + 12y + 45 = 0$.

[Oct. 2016 (D.M.E), April 2018 (D.E.C.E)]

(ii) $2x + 3y + 5 = 0$, $2x + 3y + 9 = 0$.

[Oct. 2016 (D.E.E.E)]

(iii) $2x + y + 3 = 0$, $2x + y - 2 = 0$.

(iv) $3x + 4y - 3 = 0$, $6x + 8y - 1 = 0$.

[April 2014 (A/AA/AEI/BM...)]

(v) $3x - 4y - 3 = 0$, $6x - 8y - 1 = 0$.

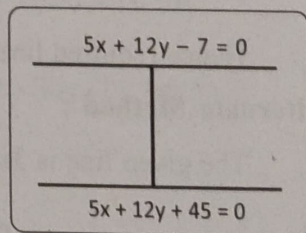
Solution :

(i) We know that the distance between the parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$

is $\left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$.

Here $a = 5$, $b = 12$, $c_1 = -7$ and $c_2 = 45$.

$\therefore$ The required distance $= \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}} = \frac{|-7 - 45|}{\sqrt{5^2 + 12^2}} = \frac{52}{13} = 4$.



(ii) Here $a = 2$, $b = 3$, $c_1 = 5$ and $c_2 = 9$.

$\therefore$ Required distance $= \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}} = \frac{|5 - 9|}{\sqrt{4 + 9}} = \frac{4}{\sqrt{13}}$

(iii) This is left as an exercise to the reader. Here required distance $= \sqrt{5}$

(iv) The given lines can be written as

$3x + 4y - 3 = 0$ and $3x + 4y - \frac{1}{2} = 0$ (Dividing by 2)

Here $a = 3$, $b = 4$, $c_1 = -3$ and $c_2 = -\frac{1}{2}$

$$\therefore \text{The distance between parallel lines} = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}} = \frac{\left| -3 + \frac{1}{2} \right|}{\sqrt{3^2 + 4^2}} = \frac{\left| -\frac{5}{2} \right|}{\sqrt{9+16}}$$

$$= \frac{5/2}{\sqrt{25}} = \frac{5/2}{5} = \frac{5}{2 \times 5} = \frac{1}{2}.$$

(v) This is left as an exercise to the reader.

Example 16:

Find the equation of the straight line passing through the point

(i) $(-3, -4)$ and parallel to the line $3x + y - 31 = 0$

[April 2018 (Common)]

(ii) $(-2, -4)$ and parallel to the line $3x + 7y + 1 = 0$.

[April 2014 (D.C.E)]

(iii) $(3, -4)$ and parallel to the line $x + 7y + 1 = 0$.

[April 2017 (A/AA/AEI...)]

(iv) $(2, 4)$ and parallel to the line $3x + y - 5 = 0$.

(v) $(-1, 2)$ and parallel to the line $x - 2y + 1 = 0$.

Solution :

(i) Required line is of the form $3x + y + k = 0$, where k is a constant (1)

Since it is passing through the point $(-3, -4)$, we have

$$3(-3) + (-4) + k = 0 \Rightarrow -9 - 4 + k = 0 \therefore k = 13 \quad \dots (2)$$

Hence required line is $3x + y + 13 = 0$ [From (1) & (2)]

Alternate Method :

The given line is $3x + y - 31 = 0$ (1)

$$\text{Its slope is } m = -\frac{\text{coeff. of } x}{\text{coeff. of } y} = -\frac{3}{1} = -3$$

Since the required line is parallel to the line (1), therefore, the slope of required line is also $m = -3$

Hence the equation of the required line passing through the point $(-3, -4)$ is

$$y - y_1 = m(x - x_1)$$

$$\text{i.e., } y - (-4) = -3[x - (-3)]$$

$$\text{or } y + 4 = -3(x + 3) = -3x - 9$$

$$\text{or } 3x + y + 4 + 9 = 0 \text{ or } 3x + y + 13 = 0.$$

(ii) This is left as an exercise to the reader. Proceeding as above, we get the required line as $3x + 7y + 34 = 0$.

(iii) This is left as an exercise to the reader.

(iv) Proceeding as above, we get the required line as $3x + y - 10 = 0$.

(v) Proceeding as above, we get the required line as $x + 7y + 25 = 0$.

Example 17:

Find the equation of the straight line passing through the point

(i) $(3, -4)$ and perpendicular to the line $3x + 5y - 21 = 0$.

[April 2015 (D.E.C.E), April 2017 (D.C.E)]

(ii) $(5, -21)$ and perpendicular to the line $3x + 5y = 0$.

(iii) $(-4, 3)$ and perpendicular to the line $x + y + 1 = 0$.

[April 2014 (D.E.C.E)]

(iv) $(3, 4)$ and perpendicular to the line $x + y + 1 = 0$.

(v) $(-4, 3)$ and perpendicular to the line $3x + y - 31 = 0$.

Solution :

(i) Equation of any line perpendicular to the straight line

$ax + by + c = 0$ is of the form $bx - ay + k = 0$.

Hence equation of the required line is of the form $5x - 3y + k = 0$, where k is a constant

Since it is passing through the point $(3, -4)$, we have $5(3) - 3(-4) + k = 0$

$$\Rightarrow 15 + 12 + k = 0 \quad \therefore k = -27$$

Thus the equation of the required line is $5x - 3y - 27 = 0$.

Alternate Method :

The given line is $3x + 5y - 21 = 0$... (1)

$$\text{Slope of line (1)} = -\frac{\text{coeff. of } x}{\text{coeff. of } y} = -\frac{3}{5} = m_1 \text{ (say)}$$

Let m_2 be the slope of the required line. Since it is perpendicular to line (1), we have

$$m_1 \cdot m_2 = -1 \Rightarrow m_2 = \frac{-1}{m_1} = \frac{-1}{-3/5} = \frac{5}{3}$$

$$\therefore \text{Slope of the required line, } m (=m_2) = \frac{5}{3}$$

Hence equation of the required line passing through the point $(3, -4)$ is

$$y - y_1 = m(x - x_1) \text{ i.e., } y - (-4) = \frac{5}{3}(x - 3)$$

$$\text{i.e., } 3(y + 4) = 5(x - 3) \text{ or } 3y + 12 = 5x - 15 \text{ or } 5x - 3y - 27 = 0.$$

(ii) Equation of the required line is of the form $5x - 3y + k = 0$

Since it is passing through the point $(5, -21)$, we have

$$5(5) - 3(-21) + k = 0 \Rightarrow 25 + 63 + k = 0 \Rightarrow k = -88$$

Hence equation of the required line is $5x - 3y - 88 = 0$.

(iii) Given line $x + y + 1 = 0$ can be written as

$$y = -x - 1 \quad \dots (1)$$

Comparing (1) with $y = mx + c$, we have

slope of the line (1) is $m_1 = -1$

$\therefore$ Slope of the line perpendicular to line (1) is

$$m_2 = \frac{-1}{m_1} = \frac{-1}{-1} = 1$$

Hence equation of the line with slope 1 and passing through the point $(-4, 3)$ is

$$y - y_1 = m(x - x_1) \Rightarrow y - 3 = 1(x + 4) = x + 4 \Rightarrow x - y + 7 = 0.$$

(iv), (v) This is left as an exercise to the reader.

Example 18:

Find the equation of the line whose perpendicular distance from the origin is 4 units and the angle which the normal makes with positive direction of x-axis is 135° .

Solution :

Here, we are given $p = 4$ and $\alpha = 135^\circ$.

$\therefore$ The equation of the line using normal form is

$$x \cos \alpha + y \sin \alpha = p$$

$$\text{i.e., } x \cos 135^\circ + y \sin 135^\circ = 4$$

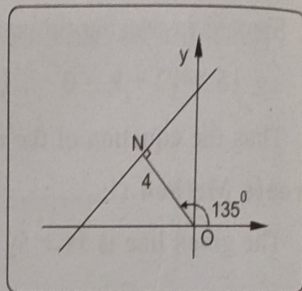
$$\text{or } x \cos (90^\circ + 45^\circ) + y \sin (90^\circ + 45^\circ) = 4$$

$$\text{or } x(-\sin 45^\circ) + y \cos 45^\circ = 4$$

$$\text{or } x\left(\frac{-1}{\sqrt{2}}\right) + y \cdot \frac{1}{\sqrt{2}} = 4$$

$$\text{or } -x + y = 4\sqrt{2} \text{ or } -x + y - 4\sqrt{2} = 0$$

$$\text{or } x - y + 4\sqrt{2} = 0.$$



Review Questions

1. (i) Write the general equation of a straight line. [Jan. 2024 (MID-I)]
(ii) Write any two forms of a straight line. [Oct. 2016 (D.E.C.E)]
(iii) List all forms of a straight line.
2. Write the equation of a straight line in slope intercept form.
3. Write the equation of the straight line passing through the points (x_1, y_1) and (x_2, y_2) . [April 2018 (D.E.E.E)]
4. Write the equation of point-slope form of a straight line. [April 2016 (D.E.E.E), April 2019 (C-165)]
5. (i) Write the intercept form of a straight line $ax + by + c = 0$. [April 2016, Oct. 2016 (D.M.E), April 2018 (D.C.E)]
(or) Write the equation of the line in intercept form. [April 2018 (Common)]
(ii) Write the normal form of the straight line.
6. Write the formula to find the angle between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$. [April 2017 (D.C.E), April 2018 (D.E.C.E)]
7. Write the conditions for two straight lines to be
(i) Parallel,
(ii) Perpendicular.
8. (i) Write the formula to find the distance between parallel lines. [Sept. 2021, Nov. 2021]
(or) Write the formula to find the distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$. [Jan. 2025 (MID-I)]
(ii) Write the formula to find the distance between the lines $2x + 3y + 9 = 0$ and $2x + 3y - 7 = 0$. [April 2018 (D.C.E)]
9. Write the formula to find the perpendicular distance from a point (x_1, y_1) to straight line $ax + by + c = 0$. [April 2017 (D.E.E.E), April 2018 (Common)]
10. Write the general form of straight line giving one example. [April 2017 (A/AA/AEI...)]
11. Write the slope of any line (i) parallel to x-axis (ii) parallel to y-axis.
12. Write the equation of the line passing through the point (x_1, y_1) and
(i) parallel to the line $ax + by + c = 0$
(ii) perpendicular to the line $ax + by + c = 0$.

Objective Questions

I. Fill in the Blanks with Correct Answers :

1. The slope of the line with $\theta = 30^\circ$ with x-axis is _____.
2. The slope of the line with $\theta = 45^\circ$ with x-axis is _____.
3. Intercepts of the line $\frac{x}{2} + \frac{y}{3} = 1$ are _____.
4. Intercepts of the line $\frac{x}{4} - \frac{y}{4} = 1$ are _____.
5. Intercepts of the line $x + 2y - 10 = 0$ are _____.
6. The lines $3x + 2y - 1 = 0$ and $2x - ky + 5 = 0$ are parallel. Then $k =$ _____.
7. Equation of a straight line in normal form is _____.
8. Perpendicular distance of the point $(1, 2)$ from the line $3x - 4y + 6 = 0$ is _____.
9. The perpendicular distance from the point $(3, 2)$ to the line $4x + y + 6 = 0$ is _____.

II. State True/False :

1. Slope of a line having an angle of inclination θ is $\cot \theta$.
2. The lines $3x + 2y + 5 = 0$ and $3x + 2y + 9 = 0$ are parallel.
3. The lines $2x + 3y - 3 = 0$ and $6x + 9y - 1 = 0$ are parallel.
4. The lines $2x - y + 5 = 0$ and $x + 2y - 8 = 0$ are parallel.
5. The distance between the parallel lines $x + 2y + 3 = 0$ and $x + 2y + 8 = 0$ is $\sqrt{5}$.
6. The point of intersection of the lines $x - y + 2 = 0$ and $2x - y + 1 = 0$ is $(1, 3)$.
7. The point of intersection of the lines $2x + 4y - 6 = 0$ and $x - 4y + 3 = 0$ is $(1, -1)$.

Answers

- | | | | | | |
|-----|-------------------------|---|------------------|---------------------------------|----------|
| I. | 1. $\frac{1}{\sqrt{3}}$ | 2. 1 | 3. 2, 3 | 4. 4, -4 | 5. 10, 5 |
| | 6. $-\frac{4}{3}$ | 7. $x \cos \theta + y \sin \theta = p, (p > 0)$ | 8. $\frac{1}{5}$ | 9. $\frac{20}{\sqrt{17}}$ units | |
| II. | 1. False | 2. True | 3. True | 4. False | 5. False |
| | 6. True | 7. False | | | |

Exercise

- Find the slope of the line passing through the points
(i) $(-a, -b)$ and (a, b) (ii) $(1, -2)$ and $(-2, 1)$
- Find the slope of the line
(i) $x + 2y - 3 = 0$ (ii) $3x - 4y + 7 = 0$
- (i) Find the value of x , if the slope of the line joining the points $(2, 5)$ and $(x, 3)$ is 2.
[April 2016 (D.E.C.E)]
(ii) If the slope of the straight line $kx + 2y + 3 = 0$ is 3, then find k .
[Oct. 2016 (D.E.E.E), April 2019 (C-16S, 2025), (C-16S), 2025]
- Find the equation of the straight line passing through the point
(i) $(1, 2)$ and having inclination $\frac{\pi}{3}$.
(ii) $(-2, -1)$ and having inclination 150° . [Jan. 2024 (MID-I)]
(iii) $(1, 2)$ and having slope 5. [April 2018 (D.E.E.E)]
- Find the equation of the line passing through the points
(i) $(1, -2), (-2, 3)$ [April 2018 (D.M.E), Jan. 2024 (MID-I)]
(ii) $(1, 2), (-3, 5)$ [April 2014 (D.M.E), 2018 (Common), Sept. 2021]
(iii) $(12, -1), (13, -4)$
(iv) $\left(\frac{3}{5}, 4\right), \left(2, -\frac{1}{2}\right)$
(v) $(at_1^2, 2at_1), (at_2^2, 2at_2)$
- (a) Find the intercepts made by the straight line
(i) $3x - 2y - 2 = 0$ [April 2015 (Common), 2017 (D.E.C.E), 2018 (D.E.E.E)]
(ii) $2x + 3y - 5 = 0$ [April 2016 (D.E.C.E), Oct. 2016 (A/AA/CH...), April 2017 (D.C.E), 2018 (D.C.M.E), April 2019 (C-16S)]
(iii) $3x + 4y - 5 = 0$. [April 2017 (D.E.E.E)]
(iv) $3x - 2y - 6 = 0$. [April 2018 (D.M.E)]
(v) $3x - 4y = 12$.
on the coordinate axes.
(b) If 3, 4 are intercepts made by a straight line on the coordinate axes, then find its equation.
[Oct. 2016 (D.E.C.E)]
(c) If 2, 3 are intercepts made by the straight line then find its equation. [April 2024]

7. Find the angle between the lines

(i) $2x - y + 3 = 0$ and $x + y - 2 = 0$ [April 2018 (D.M.E)]

(ii) $x + 2y + 9 = 0$ and $3x + y - 7 = 0$ [April 2014 (D.CM.E)]

(iii) $x + y + 1 = 0$, $x = 5$.

(iv) $2x + y + 4 = 0$ and $y - 3x = 7$. [April 2018 (D.C.E, Common), Jan. 2025 (MID-I)]

(v) $x + 3y - 1 = 0$ and $2x + 5y - 2 = 0$ [Oct. 2015 (D.E.C.E), April 2017 (D.E.C.E, D.M.E), Nov. 2021 (Sup)]

(vi) $5x + 2y + 11 = 0$ and $5x - 3y + 9 = 0$

8. If the acute angle between the lines $3x - y + 4 = 0$ and $kx + y + 2 = 0$ is $\frac{\pi}{4}$, find value of k .

9. (a) Find the perpendicular distance of the point

(i) $(3, -5)$ from the line $3x - 4y - 26 = 0$ [Oct. 2010 (D.CM.E), April 2016 (D.E.E.E), 2017 (D.M.E), April 2018 (D.E.C.E)]

(ii) $(2, 4)$ from the line $4x - 3y - 6 = 0$. [April 2018 (D.CM.E), Jan. 2025 (MID-I)]

(iii) $(2, 3)$ from the line $2x + 3y + 5 = 0$

(iv) $(-2, 3)$ to the line $x - 7y + 15 = 0$.

(v) Origin i.e., $(0, 0)$ to the line $3x - 4y - 20 = 0$. [April 2017 (D.E.C.E), 2018 (D.M.E), April 2019 (C-16S)]

(vi) $(3, -2)$ to the line $3x - 4y - 10 = 0$.

(vii) $(-2, 3)$ to the line $3x - 4y - 20 = 0$.

(b) Find the length of the perpendicular from the origin upon the straight line $\frac{x}{a} + \frac{y}{b} = 1$

10. Find the distance between the parallel lines

(i) (a) $3x + 4y - 6 = 0$ and $3x + 4y + 2 = 0$ [Oct. 2015 (D.M.E, D.E.E.E), April 2018 (D.M.E)]

(b) $3x + 4y + 5 = 0$ and $3x + 4y + 8 = 0$.

(ii) (a) $3x - 4y + 7 = 0$ and $3x - 4y + 5 = 0$ [April 2018 (D.CM.E)]

(b) $3x - 4y + 7 = 0$ and $3x - 4y + 4 = 0$. [April 2018 (D.E.E.E)]

(iii) $3x - 4y - 8 = 0$ and $6x - 8y + 5 = 0$.

(iv) $3x + 2y - 9 = 0$ and $3x + 2y + 12 = 0$. [April 2015 (D.M.E), Jan. 2024 (MID-I)]

(v) $4x - 3y + 9 = 0$ and $4x - 3y + 4 = 0$. [April 2017 (A/AA/AEI...)]

(vi) $3x + 4y = 9$ and $6x + 8y = 15$.

11. (i) Find the equation of the straight line passing through the point $(3, -4)$ and
- (a) parallel to the line $3x + 5y - 21 = 0$.
 - (b) perpendicular to the line $3x + 5y - 21 = 0$.
 - (c) perpendicular to the line $5x + 3y + 1 = 0$.
- (ii) Find the equation of the line parallel to the line $3x - 4y + 2 = 0$ and passing through the point $(-2, 3)$.
- (iii) Find the equation of the straight line passing through the point
- (a) $(1, 2)$ and parallel to the line $3x + 4y - 6 = 0$.
 - (b) $(3, -5)$ and parallel to the line $x - 7y + 15 = 0$.
 - (c) $(3, -7)$ and parallel to the line $2x + 3y + 5 = 0$.
12. (i) Find the equation of the straight line passing through the point $(3, -4)$ and perpendicular to the line $5x + 3y - 1 = 0$ [April 2017 (D.E.E.E), 2018 (D.E.C.E)]
- (ii) Find the equation of the straight line passing through the point $(2, -5)$ and perpendicular to the line $7x + 2y - 1 = 0$. [April 2014]
- (iii) Find the equation of the straight line perpendicular to the line $2x + y + 4 = 0$ and passing through the point $(3, 4)$. [April 2018 (D.E.E.E)]
- (iv) Find equation of straight line passing through the point $(3, 4)$ and perpendicular to the line $3x + 5y - 21 = 0$. [April 2018 (D.C.M.E)]
- (v) Find the equation of a line perpendicular to the line $x - 2y + 3 = 0$ and passing through the point $(1, -2)$.
13. Find the point of intersection of the lines
- (i) $x - 3y + 6 = 0$, $2x + 3y - 10 = 0$. [April 2012 (D.E.C.E), 2018 (D.C.E), 2019 (C-16S)]
 - (ii) $2x + 4y = 6$ and $x - 4y = -3$.
 - (iii) $5x - 7y + 1 = 0$ and $2x + 5y - 11 = 0$.
 - (iv) $x + 3y - 6 = 0$ and $y - 3x = 7$.
14. Find the equation of the line whose perpendicular distance from the origin is
- (i) 4 units and the angle which the normal makes with positive direction of x-axis is 15° .
 - (ii) 5 units and the angle which the normal makes with positive direction of x-axis is 60° .
 - (iii) 7 units and its slope is -1 . Find the equation of the line.

Answers

1. (i) $\frac{b}{a}$ (ii) -1
2. (i) $-\frac{1}{2}$ (ii) $\frac{3}{4}$
3. (i) 1 (ii) -6
4. (i) $\sqrt{3}x - y + 2 - \sqrt{3} = 0$ (ii) $x + \sqrt{3}y + \sqrt{3} + 2 = 0$ (iii) $5x - y - 3 = 0$
5. (i) $5x + 3y + 1 = 0$
 (ii) $3x + 4y - 11 = 0$
 (iii) $3x + y - 35 = 0$
 (iv) $45x + 14y - 83 = 0$
 (v) $y(t_1 + t_2) = 2x + 2a t_1 t_2$
6. (a) (i) $\frac{2}{3}, -1$ (ii) $\frac{5}{2}, \frac{5}{3}$ (iii) $\frac{5}{3}, \frac{5}{4}$ (iv) $2, -3$ (v) $4, -3$
 (b) $4x + 3y - 12 = 0$ (c) $\frac{x}{2} + \frac{y}{3} = 1$ (or) $3x + 2y - 6 = 0$
7. (i) $\tan^{-1}3$ (ii) $\frac{\pi}{4}$ (iii) $\frac{\pi}{4}$ (iv) $\frac{\pi}{4}$
 (v) $\cos^{-1}\left(\frac{17}{\sqrt{290}}\right)$ (or) $\tan^{-1}\left(\frac{25}{19}\right)$ (vi) $\cos^{-1}\left(\frac{19}{\sqrt{986}}\right)$
8. $-\frac{1}{2}$ or 2
9. (a) (i) $\frac{3}{5}$ (ii) 2 (iii) $\frac{18}{\sqrt{13}}$ (iv) $\frac{8}{5\sqrt{2}}$ (v) 4 (vi) $\frac{7}{5}$ (vii) $\frac{38}{5}$
 (b) $\frac{|ab|}{\sqrt{a^2 + b^2}}$
10. (i) (a) $\frac{8}{5}$ (b) $\frac{3}{5}$ (ii) (a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (iii) $\frac{21}{10}$ (iv) $\frac{21}{\sqrt{13}}$
 (v) 1 (vi) $\frac{3}{10}$
11. (i) (a) $3x + 5y + 11 = 0$ (b) $5x - 3y - 27 = 0$ (c) $3x - 5y - 29 = 0$
 (ii) $3x - 4y + 18 = 0$ (iii) (a) $3x + 4y - 11 = 0$ (b) $x - 7y - 38 = 0$
12. (i) $3x - 5y - 29 = 0$ (ii) $2x - 7y - 39 = 0$ (iii) $x - 2y + 5 = 0$
 (iv) $5x - 3y - 3 = 0$ (v) $2x + y = 0$
13. (i) $\left(\frac{4}{3}, \frac{22}{9}\right)$ (ii) $(1, 1)$ (iii) $\left(\frac{72}{39}, \frac{57}{39}\right)$
14. (i) $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$ (ii) $x + \sqrt{3}y - 10 = 0$ (iii) $x - y + 7\sqrt{2} = 0$